



Evaluation of LLMs in Medical Text Summarization: The Role of Vocabulary Adaptation in High OOV Settings

Gunjan Balde, Soumyadeep Roy, Mainack Mondal, and Niloy Ganguly

Indian Institute of Technology Kharagpur

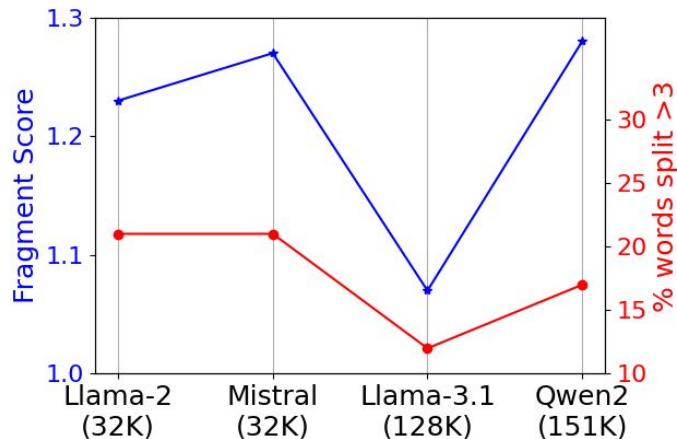
ACL 2025
VIENNA 

LLMs efficient in processing open-domain text

While processing open-domain text
(CNN-Dailymail news articles):

- a) on average a word is split into **1.18** subwords
- b) 18% of words in the datasets are split more than thrice

CNN Dataset: Fragment Score and Split Fraction

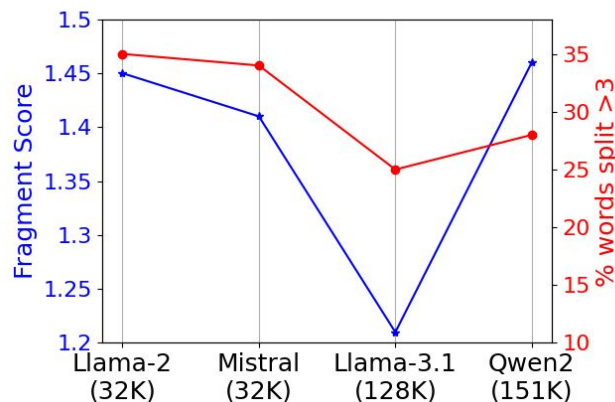


But over-fragments medical domain text

However, when processing medical text (PubMed abstracts):

- a. on average a word in dataset is split into **1.38** subwords (increase of 17%)
- b. 30% words in dataset split more than thrice (increase of 67%)

PAC Dataset: Fragment Score and Split Fraction



So how does generalist LLMs perform in medical summarization tasks?

Benchmarking LLM on medical summarization

Recent efforts (like ClinSumm^[1]) have benchmarked LLMs for task of medical summarization using ICL and QLoRA-based fine-tuning techniques

Such benchmarking efforts consider only **whole dataset for evaluation** and **miss fine-grained evaluation scenarios**

Evaluation under such scenarios truly test LLMs generalizability capabilities

Evaluation in challenging fine-grained scenarios

Scenario-1: When source document and the reference summaries have a high concentration of medical words which are split more than X {1,3}—***high OOV (Out-of-Vocabulary) concentration***

Scenario-2: When the reference summary has high concentration of unigrams that are not present in the source document — ***high novelty concentration***

RQ1: How do LLMs perform in such challenging fine-grained scenarios of high OOV and Novelty?

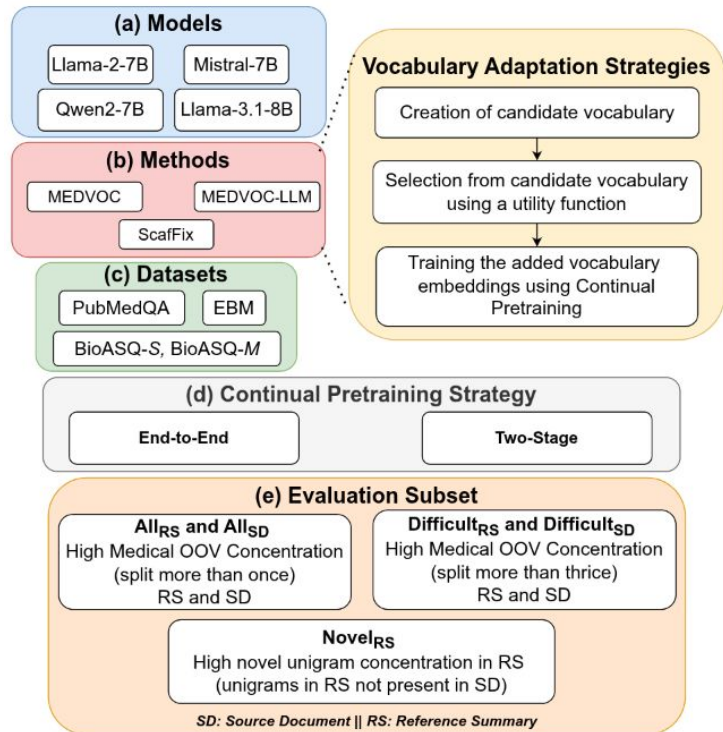
RQ1: How do LLMs perform in such challenging fine-grained scenarios of high OOV and Novelty?

RQ2: Can Vocabulary Adaptation help improve overall performance in challenging scenarios?

Benchmarking setup for fine-grained evaluation

In our study, we benchmark

- (a) four recent mainstream SoTA models;
- (b) on three vocabulary adaptation methods;
- (c) using three medical dataset;
- (d) over two continual pretraining strategy;
- (e) in scenarios on OOV and Novel unigram



Vocabulary adaptation techniques considered

MEDVOC^[1]

- MEDVOC adapts PLM vocabulary during fine-tuning stage
- Vocabulary obtained from Medical OOV words in downstream task and large-scale in-domain dataset (*PAC*)
- Optimal-sized vocabulary set identified by optimizing fragment score

Vocabulary adaptation techniques considered

MEDVOC-LLM— Mitigating issues in MEDVOC

- Some tokens added in MEDVOC are not useful in generation
 - Several tokens populated from PAC infrequent in downstream task dataset
 - Some added tokens mixture of numerals, punctuations, and letters
 - Eg., _95%, =1. etc —non-standard for LLM tokenizer
- We remove such tokens from MEDVOC vocabularies and add only the filtered set

Vocabulary adaptation techniques considered

ScaffFix— Mitigating the need of intermediate tokens

- In order to add a medical word in the vocabulary of LLM,
 - the addition procedure has to also add its subwords in an incremental fashion
 - Introduce terms which are infrequent and rarely used after longer word addition
—*Scaffolding tokens*^[1,2]
- Leveraging AdaptBPE^[3], only whole medical are added — getting rid of intermediate scaffolding tokens

[1] Lian et al. 2025. Scaffold-bpe: Enhancing byte pair encoding for large language models with simple and effective scaffold token removal. AAAI

[2] Cognetta et al. 2024. [An analysis of BPE vocabulary trimming in neural machine translation](#). Fifth Workshop on Insights from Negative Results in NLP (ACL)

[3] Balde et al. 2024. Adaptive BPE tokenization for enhanced vocabulary adaptation in finetuning pretrained language models. EMNLP (Findings)

Fine-Tuning LLM with adapted vocabulary

- Task: **NTP (Next Token Prediction)** task in an autoregressive fashion
- Embedding initialized with the average of the existing subword

$$EMBED_{\text{cholesterol}} := \text{AVG}[Embed_{\text{ch}}, EMBED_{\text{ol}}, EMBED_{\text{ester}}, EMBED_{\text{ol}}]$$

Fine-Tuning LLM with adapted vocabulary

Continual pretraining on **20K PubMed Abstracts** using LoRA:

- **End-to-End:** The model is trained in an end-to-end manner by freezing all the base model layers except the input and output embedding layers and training LoRA adapters.
- **Two-Stage:** First, the entire model along with the LoRA layers is frozen except for the input and output embedding layers for a short duration, then unfreeze the LoRA adapters and train the LoRA adapters along with the embedding layers

Experimental setup: Models, Dataset, and Metrics

- **LLMs**

- Llama-2-7B; Mistral-7B (Vocab size: 32K vocab size)
- Llama-3.1-8B; Qwen2-7B (Vocab Size: 128K and 151K)

- **Datasets:**

- EBM, BioASQ, PubMedQA
 - Source Document (SD) —Query + PubMed Abstract
 - Reference Summary (RS) —Answer

- **Metrics:** Rouge (R-LCS) and Concept-Score^[1] (CSr)

Experimental setup: Baselines

- **BASE.** BASE corresponds to the base variant of original LLM without any vocabulary adaptation and continual pretraining
- **CPT-Only.** Original BASE LLM that has undergone the continual pretraining (CPT) but without any vocabulary adaptation

Challenging scenarios considered

Categories of Words	
Difficult-OOV	Medical words that are split more than thrice by model tokenizer
Novel	Words in the summary that are not present in the source document

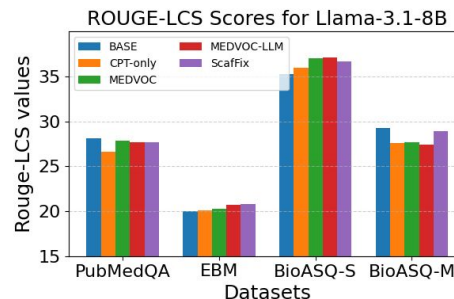
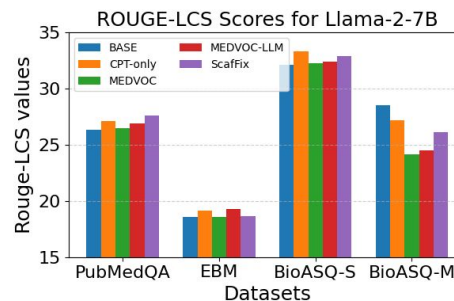
Evaluation Setting (Top ten percentile)	
Difficult _{RS}	High Difficult-OOV concentration in the reference summaries
Difficult _{SD}	High Difficult-OOV concentration in the source document
Novel _{RS}	High Novel concentration in the reference summaries

Sample Difficult-OOV words

Word	Llama-3.1-8B (128K)
hypercholesterolemia	'Ġhyper', 'cho', 'le', 'ster', 'olem', 'ia'
antipyretics	'Ġant', 'ipy', 'ret', 'ics'
Osteoarthritis	'ĠO', 'ste', 'o', 'ar', 'thritis'

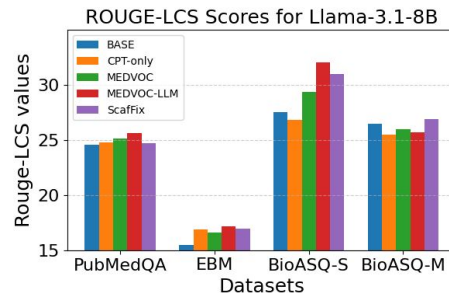
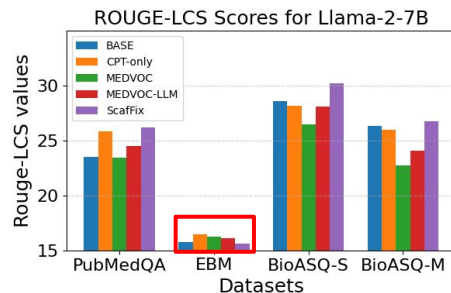
Result-1: Vocabulary adaptation outperform BASE model on full data

- At least one of the vocabulary adaptation strategies outperforms BASE in **5/8** settings
- **ScaFix**, is the best-performing vocabulary adaptation method



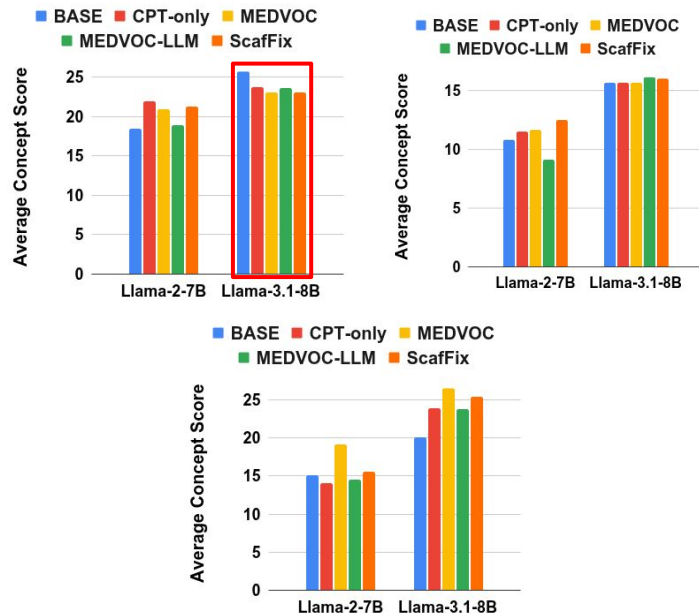
Result-2: Vocabulary adaptation helps in challenging scenarios

- We observe that vocabulary adaptation outperforms BASE in **7/8** settings
 - Average of $Difficult_{RS}$, $Difficult_{SD}$, $Novel_{RS}$
- **ScaFix**, is the best vocabulary adaptation strategy in such settings



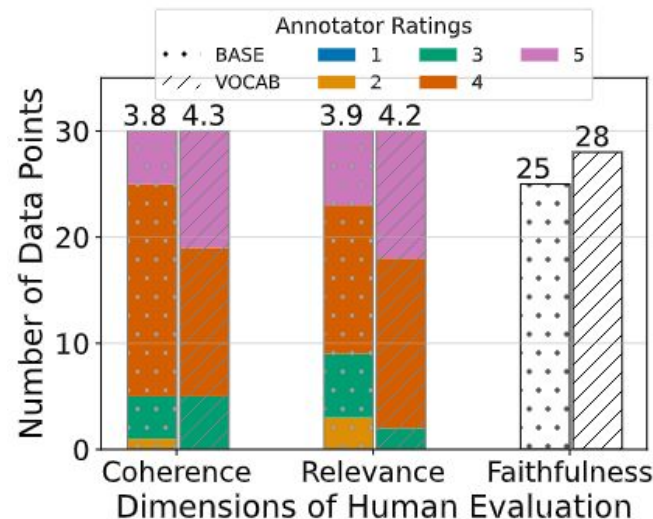
Result-3: Vocabulary adaptation improves factual consistency over BASE

- At least one vocabulary adaptation strategy improves over BASE in **5/6** settings
- The overall average improvement is 18.75% for Llama-2 and 14.82% for Llama-3.1



Result-4: Human evaluation favours vocabulary adaptation

- 30 data points (BASE v/s ScafFix) with high difficult OOV concentration
- Annotated across three aspects: Relevance, Coherence (on a Likert scale of 1 to 5), and Faithfulness (binary) by medical experts
- **ScafFix** generates more faithful and more relevant summaries



Conclusion and Key Takeaways

- We first show over-fragmentation is a persistent issues even for higher vocabulary size LLMs
- We then identify the need to benchmark LLMs in challenging scenarios
- We show vocabulary adaptation improves over BASE in challenging scenarios
- Human evaluation find summaries generated by vocabulary adapted models to be more relevant, coherent, and faithful

Thank You!



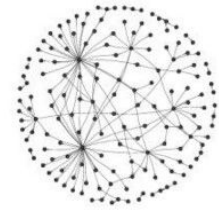
Soumyadeep Roy



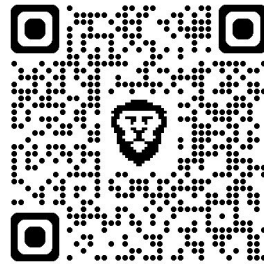
Mainack Mondal



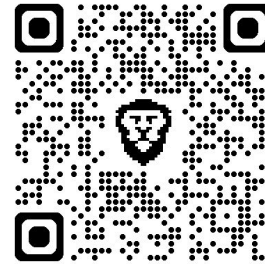
Niloy Ganguly



CNeRG



Codebase



Preprint